

Requirements Table – Second-Hand Product Escrow Marketplace

Problem Statement #39

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system would lock buyer payment in escrow and the transition order state to 'Under Inspection' until a inspector submits physical verification sign-off.	High	Pass: Funds held securely and released only after dual verification. Fails if Funds released prematurely to seller.	Protects the buyer funds by guaranteeing independent verification before transfer to the seller (given requirement).
FR-002	Functional	The system shall allow a seller to list a secondhand product with a condition description, category, price, and at least one photo before listing becomes visible in search results.	High	Pass: A listing submitted with all mandatory fields (description, category, price, ≥ 1 photo) appears in search within 5 seconds; an incomplete submission is rejected with a field-level error. Fail: An incomplete listing is published, or the search index does not update within 5 seconds.	Ensures that the buyers have enough accurate, verifiable information to decide on a purchase, reducing disputes over item condition later.
FR-003	Functional	The system shall notify the buyer within 60 seconds of an inspection agent submitting a physical verification report.	Medium	Pass: In a test order, the buyer receives an in-app and email notification containing the report status within 60 seconds of submission . Fail: Notification is late, missing, or omits the report status.	Keeps the buyer promptly informed so that the mandatory approval window can begin without avoidable delay.
FR-004	Functional	The system shall require the buyer to approve or reject an inspected product within a 48-hour review window after the inspection report is posted; unactioned orders shall be automatically flagged for admin review.	High	Pass: An order left unactioned 48 hours after report posting is automatically added to the admin review queue with a timestamp. Fail: The order is not flagged after 48 hours, or the buyer cannot record a decision within the window.	Prevents indefinite escrow lock-up and guarantees a timely fund release or dispute initiation.
FR-005	Functional	The system shall allow a buyer to file a dispute against an inspection report or a delivered product, which shall immediately transition the order to 'Disputed' status and suspend any scheduled automatic funds release.	High	Pass: Submitting a dispute changes the order status to 'Disputed' within 2 seconds and no scheduled fund release executes while the status is 'Disputed'. Fail: A fund release proceeds after a dispute is filed, or the status fails to change.	Gives buyers formal recourse and guarantees that escrow funds stay protected while an independent claim is investigated.

Nonfunctional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Dispute resolution tickets must freeze escrow funds immediately and prevent the account withdrawals during active investigation.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load with 100% of active dispute accounts blocked from withdrawal. Fail: Any withdrawal succeeds, or the freeze is not applied within the target latency, during an active investigation.	Prevents fund tampering or dissipation while a dispute is unresolved, preserving integrity of the escrow guarantee (given requirement).
NFR-002	Security	The system shall encrypt all escrow, payment, and personal identification data both at rest and in transit, using industry standard encryption (e.g., AES-256 at rest, TLS 1.2+ in transit).	High	Pass: A security scan confirms all escrow/payment/PII data stores use AES-256 encryption at rest and all API endpoints reject connections below TLS 1.2. Fail: Any escrow, payment, or PII data is found unencrypted at rest, or transmitted over an unencrypted or sub-TLS-1.2 channel.	Protects sensitive financial and personal data from breach, given the high-value, trust-sensitive nature of escrow transactions.